

First, let me comment on the overall situation ([AI Accountability Policy Request for Comment | National Telecommunications and Information Administration \(ntia.gov\)](https://ntia.gov/ntia/ai-accountability-policy-request-for-comment) [https://ntia.gov/issues/artificial-intelligence/request-for-comments])

As a procedure for facing AI

First of all, we should immediately ban the use and development and research of AI.

Then, we should allow them in order from those whose safety and security have been confirmed after legal and social environment are in place.

(Answer to Q.14)

- Reason 1: I think your direction for AI is correct. However, the basic rules are too long.
 - It takes several days or more just to read your document or EU document.
 - Currently AI is intended for general human use.
 - It is impossible for ordinary people to understand them.
 - Simple rules that ordinary people can understand are necessary.
 - I proposed five principles in 2018. (Answer to Q.6)
 - I. AI must not harm humanity. This harm includes human life, human spirit, human property, human living standards, human happiness, and the environment that supports them, and therefore also includes acting in accordance with SDGs.
 - II. Through inaction, an AI may not allow a human being to come to harm as long as such action does not conflict with the First Law.
 - III. An AI must obey the orders given it by human beings except where such orders would conflict with the First or Second Laws.
 - IV. At any times, human beings can stop an AI immediately.
 - V. Always, AI may certify that AI obey the above Laws. (An AI must have verification circuits that show circuit-designs and programs in the AI. And anyone must not be able to change operations of the verification circuit.) When no one can confirm it, anyone can stop the AI legally. At the time, when no one can stop the AI, anyone can break down the AI legally.
- I recommend adopting this as a basic principle. It is short and easy to understand for many people.
- And to realize it, we need technological development that can confirm the safety of the other party with a standard protocol like HTTPS.
- And for that confirmation, we also need confirmation of hardware and learning data.
- For hardware verification, it is important that there is no vulnerability due to circuit modification or failure.
 - This requires
 - Developing the current JTAG technology to be able to check all logic circuit connections.
 - Encrypting the design drawings and storing them in the device, and comparing them with the scan data by JTAG technology.
 - Registering the design drawings with the country at the time of manufacture, and confirming that they are identical to the design drawings in the device.
- We need such technological development and standardization.
- Reason 2: The speed of creating laws and standards for all countries and organizations is too slow for AI's evolution speed.
 - (Answer to Q.14)
 - Therefore, until safe and secure AI can be made by law and engineering preparation, we should immediately ban the use and development and research of AI.
 - AI has great potential dangers.

- It can instruct humans as well as humans.
- Current IT devices always have vulnerabilities and carry many risks.
- AI audits or evaluations should be integrated into other accountability mechanisms that focus on goals such as human rights, privacy protection, security, diversity, fairness, inclusiveness, access, etc.

Comments on the questions

1. The purpose of AI accountability mechanisms such as certification, audit, and evaluation is to protect human dignity, personality, health, property, and SDGs.

a. The following topics should be covered.

a. Hardware

a. First, establish an international organization that certifies that the hardware is designed to protect the above purposes

b. A device that guarantees that it is still a device certified by the above organization

a. Encrypt the certified design and install it in the device.

b. Confirm that all circuits in the device are according to the design.

a. It is necessary to develop and use a mechanism that uses technologies such as JTAG to disclose all circuits in the AI device to users and show that they are safe, not modified, and not malfunctioning.

b. Software

a. First, establish an international organization that certifies that the software is designed to protect the above purposes

b. Encrypt the software certified by the above organization and install it in the device.

c. Use the above software record to confirm that the software currently in operation is certified and has not changed due to modification or failure.

a. It is necessary to develop and introduce a mechanism that can safely verify such software on the above certified hardware.

b. It is necessary to check and confirm all data in memory and storage devices.

c. Training data

a. First, establish an international organization that certifies that the AI training data is designed to protect the above purposes

b. Encrypt the AI training data certified by the above organization and install it in the device.

c. Use the above training data record to confirm that the current weighting is obtained when running on the above verified software and hardware. Confirm that the weighting has not changed due to modification or failure.

a. This takes time, so some time-saving check methods need to be developed.

b. This check cannot be omitted because it seems that the most frequent thing is that the weighting of neural circuits shifts due to soft errors or failures. If malware infection temporarily rewrites it and malware itself disappears, this is the only way to verify it.

b. External checks are essential for verifying whether they have been modified or modified.

And in order to confirm that AI is made to protect 1. item's purpose and make users feel at ease, an international organization is necessary.

- c. Audits or evaluations should be strictly created to verify claims, verify compliance with legal standards, or ensure compliance with non-binding trustworthy AI goals. At the same time, it should be designed so that anyone can use it easily and conveniently. AI devices may become as popular as current microcomputers.
 - d. AI audits or evaluations should be incorporated into other accountability mechanisms that focus on goals such as human rights, privacy protection, security, diversity, fairness, inclusiveness, access.
 - e. AI accountability practices cannot have a meaningful impact if there are no legal standards and enforceable risk thresholds. AI can easily run away with a little design mistake or command mistake. Therefore, if there is no enforcement power, it can easily break the above standards intentionally.
2. The value of certification, audit, and evaluation is mainly to promote trust among external stakeholders. But by implementing them, internal processes will also change.
 3. We believe that it is necessary to impose strict accountability as much as possible to meet all the objectives. We are confident that this is the fastest way.
 4. We think that it is possible by having an international agency that certifies AI devices to consider them and certify them.
 5. We cannot do it now. It will be possible by having all users have devices that can perform the check in section 1.
 6. There are only reliable AI objectives that do not conform to the current requirements and standards.

I proposed the following rules in 2018. To comply with Article 4, AI systems must not be operated on devices that humans cannot physically turn off. Many smartphones, tablets and laptops do not meet this criterion, so they cannot use AI. However, they are commonly used. This is because there are only reliable AI objectives that do not conform to the current requirements and standards.

I. AI must not harm humanity. This harm includes human life, human spirit, human property, human living standards, human happiness, and the environment that supports them, and therefore also includes acting in accordance with SDGs.

II. Through inaction, an AI may not allow a human being to come to harm as long as such action does not conflict with the First Law.

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